

NAME

ppqac - exoteric and esoteric notes on post-quantum access control

SYNOPSIS

ppqac is a short paper, not a utility. Read it as `man(7)`, or render a PDF:

```
mandoc -T pdf ppqac.7 > ppqac.pdf
```

ABSTRACT

A quantum adversary is not a quantum operating system. The defender considered here is still a von Neumann machine: binary words, a shared store, a kernel that fetches what it polices. The adversary is allowed *BQP* and Grover search, and -- as a working hypothesis -- coupling to the electrical plant of that machine by entanglement or entanglement-enhanced sensing.

The *exoteric* claim is narrow and already usable: discretionary access control, mandatory access control, and MMU isolation do not rest on a problem in *BQP*. They are quantum-adversary-stable *mediation* of a classical reference monitor. What remains post-quantum-sensitive is *naming* (who the subject is before the label attaches): `sshd(8)`, TLS, package signatures.

The *esoteric* claim is the same electrons. A page-table walk is the power rail. A label in an extended attribute is a pattern of wells that, when read, wiggles a bitline. In-band and out-of-band were a filing convenience. If a probe can taste switching, it is tasting mediation. Containing who may bring a high object into a core is then a lid on which recipes may boil -- resistance, not a Faraday cage, and not a proof that quantum coupling is cheap.

This suite's practice (`mac_grok(8)`, `mac_lomac_grok(8)`) is the exoteric half made operational on FreeBSD. The esoteric half is philosophy the operator may yet prove. The paper states both, and the obligations of a proof, without pretending the second half is already a theorem.

THREAT MODEL**Defender**

Classical stored-program host. FreeBSD GENERIC, `mac(4)` available, `mac_lomac(4)` as the integrity policy, Unix DAC still in force, MMU and virtual memory in force. Not a quantum ISA. Not a promise that the kernel is small.

Adversary

Quantum. Polynomial-time quantum algorithms (Shor on the usual public-key names) and quadratic unstructured search (Grover). May also attempt physical coupling to power, ground, and EM (the plant) including, by hypothesis, entanglement or squeezed sensing. Does not get a free write to kernel memory

from having qubits in a van.

Out of scope as already solved

Replacing RSA with ML-KEM or ML-DSA. That is post-quantum *cryptography*. It is necessary on the naming path and insufficient as mediation.

EXOTERIC

This is the face you can teach from the Handbook in one sitting.

Naming versus mediation

Cryptography binds a name to a subject: host keys, user keys, package signatures, TLS to a mirror. Access control binds a subject already named to objects and to regions of the store: mode bits, MAC grades, page rights.

Shor wrecks the usual naming path. It does not mint ‘lomac/high’ and it does not allocate a PTE. A BQP adversary who wishes to *act as* a subject still has to produce a classical write the vnode path, the MAC hook, and the MMU will accept. That is the sense in which DAC + MAC + MMU are **quantum-adversary-stable isolation**: they never took a hardness assumption in number theory.

What does not rest on BQP

- ✦ Unix DAC: `chmod(2)`, `chown(2)`, `uid/gid`. Owner discretion. If it fails, the owner (or a hijacked uid) grants the world.
- ✦ `mac_lomac(4)`: integrity, low-watermark. Read low, demote, cannot write high. The comparison is ‘subject::higrade $\geq$ object::grade’. The same comparison at 1 Hz and at 100 GbE. Extra bandwidth demotes or returns EPERM sooner.
- ✦ The MMU and VM subsystem: address spaces, W^X , rings. `mac_lomac(4)` may revoke `mmap(2)` on demotion. The MMU does not know a LOMAC grade; the policy uses the VM machinery.
- ✦ Optional orthogonal modules: `mac_seetheruids(4)` and friends. Hide, silence, port-bind. Still classical hooks.

These layers are *not* semi-redundant copies of one lock. They fail differently. A `chmod(1)` the owner should not have granted does not punch a high label. A wild pointer does not rewrite `/etc/login.conf`. A demoted process still cannot write `/bin` even if DAC would allow it. That is mechanism diversity against a quantum adversary who only breaks *one* kind of thing (a modulus, a mapping, a mode bit).

What remains fragile

Whoever the quantum adversary *becomes* still has to pass `sshd(8)` (or equivalent) before `login.conf(5)`

attaches 'lomac/...'. Isolation after login is the part this suite actually builds. The naming path is a separate retrofit (post-quantum public key, or a non-cryptographic gate). Do not advertise 'root contained' (mac(9)) and do not advertise 'Kyber by kldload'.

Practice

mac_grok(8) snapshots the pool (zfs(8) **snapshot -r**) on filesystems and zvols, optional zpool-checkpoint(8), then stages mac_lomac_grok(8) and mac_generic_grok(8). Enforce only in a test window. Walk Handbook ch.19 troubleshooting and the package *README* section "Known issues" first. Uninstall restores pre-install *behavior* from *PREINSTALL*; it is not a forensic wipe and it is not a pool rewind.

LOMAC is chosen because the proof is short: a sandbox subject cannot write-up into high OS objects after a low read. Roles are pw(8) groups bound to login.conf(5) classes, not per-account specfiles. That is the exoteric work.

ESOTERIC

This is the face you tell someone who has sat with the die and is willing to mix the metaphors until they share a pot.

Same electrons

The MMU is not a fairy standing beside the power distribution network. A PTE walk is the rail. chmod(2) is charge in a cache line. A LOMAC label in a ZFS system extended attribute is a pattern of wells that will, when interpreted, wiggle a bitline.

"In-band" and "out-of-band" were a clerk's distinction: the kernel's story about the soup, and the soup. If a probe can taste switching, it is tasting *mediation*. Containing who may bring a high object into a core is then not "software, therefore irrelevant to physics". It is putting a lid on which recipes are allowed to boil.

Three shadows, one body

DAC, MAC, MMU: on the whiteboard, orthogonal. In the die, one animal's shadows on three walls. Owner bit, grade, page right -- three names for "this subject may disturb this region of the store".

A quantum adversary who only factors moduli walks around the animal. One who couples to the store is fighting the animal. Semi-redundant, if you must use that word: not two padlocks on one hasp, but overlapping umbra. Break one story and the other two still describe the same current.

The tide and the cup

mac_lomac(4) is a tide-line. Read low, you sink, you cannot write the hill. A quantum adversary is not a bigger wave in that metaphor; Shor is a different ocean. Grover is a pump with a square-root crank. The tide-line does not move because the crank got better.

von Neumann's cup: code and data drink together. The MMU, NX, rings, MAC -- partitions in the cup. A quantum ISA would smash the cup. A quantum *adversary* still has to sip from *this* cup. The claim "entanglement with the PSU" is the claim that you can sip without using the handles we installed. The handles were never "not electricity". They were *named* electricity. Naming is cheap. Coupling to a named net in a 3 GHz kitchen is not.

Probe as guest, not ghost

Shor is the celebrity quantum trick: clean algebra, cold qubits, a problem written down in advance. Coupling usefully to a wall-wart, or even Heisenberg-limited listening to one rail under a ground plane, is the trade. Thermal baths eat Bell pairs for breakfast. Squeezed sensors help signal-to-noise; they do not print a map of 'lomap/high'. You still need placement, bandwidth, a correlator, and usually a local modulator. That modulator is a process. Processes have uids, labels, mappings.

Do not presume the quantum things come easy. Factoring is the paper. Coupling is the apprenticeship.

Named and unnamed stock

If the channel factors through a kernel-visible subject, the monitor still bites: sandbox, LOMAC, seeotheruids, page rights constrain who may *play* a secret onto the bus for a probe to hear (the RAMBO-shaped conductor, now with a fancier microphone). That is resistance.

Imperviousness of the electrical plant is a different soup. It needs a no-distinguishability claim: for every allowed run, the adversary's observable is independent of high objects. POSIX MAC does not give you that. Constant-power logic, filtered PDN, shielding, not cooking secrets in a hot core -- those are unnamed stock. You may compose them with MAC (high work only in a shielded domain). You cannot derive them from *mac_lomac.ko*.

Integrity is still the wrong primary label for leakage. LOMAC lets you read down and demotes you. An authorized trusted subject computing on a high object *is* a hot core. Confidentiality-style non-interference is the property a leakage thesis must prove; demotion-on-read is a lemma at best, a mute, not a cage.

THESES

Numbered so they can be denied separately.

T1

A quantum adversary does not obtain a free classical write from BQP alone. The reference monitor still runs.

T2

DAC, MAC, and MMU are quantum-adversary-stable mediation on a von Neumann defender. They are post-quantum *access control* in that sense, and not post-quantum cryptography.

T3

(working hypothesis) Coupling to the classical electrical plant, including by entanglement or entanglement-enhanced sensing, can exist as an attack channel.

T4

If that channel factors through a kernel-visible subject or object, T1 and T2 still apply: the cook can be labeled. If it does not factor, T1 and T2 are silent, and you are in TEMPEST and enclave land.

T5

Quantum advantage against the plant does not come easy. Decoherence, placement, and distinguishing a vnode in GENERIC's noise are the adversary's proof burden, not a lemma from Shor.

T1 and T2 are the exoteric paper, and this suite implements them. T3 is presumed so T4 can be stated. T4 is two theorems, not one, until the coupling map is on the page. T5 is the discipline that keeps T3 from swallowing the work.

PROOF OBLIGATIONS

A proof of the esoteric half, if the operator can finish it, must:

1. Exhibit a coupling map from labeled computation to an observable the quantum adversary can distinguish (current, B-field, impedance echo, squeezed magnetometry).
2. Show whether that map factors through a kernel-visible subject (process, credential, mapping, vnode) or through the plant alone (authorized hot core, or idle rail).
3. If it factors, show that the policy (DAC + MAC + MMU) makes the observable independent of what one cares about, or bound the remaining leakage. Independence is imperviousness. A bound is resistance. Do not use one word for the other.
4. If it does not factor, say so, and move the remaining controls out of mac(4) into physics: shielding, PDN filter, constant power, air gap.
5. Keep LOMAC's actual axiom. Do not smuggle no-read-down out of a low-watermark. If the thesis is secrecy of emanations, say MLS or Biba or an enclave, and say why this suite is still the lid on the cook.

DO NOT PRESUME

- ☛ Do not presume that "post-quantum" will be read as anything but Kyber unless you say **pqac** aloud.
- ☛ Do not presume a quantum ISA. The cup is still binary.

- ⚡ Do not presume Grover finds kernel bugs. Unstructured search is not a confused-deputy oracle.
- ⚡ Do not presume entanglement with a 300 K PSU is free, nor that it is impossible. State T3, then pay for T4.
- ⚡ Do not presume `kldload(8)` of **mac_lomac** is a Faraday cage, a naming retrofit, or a proof. It is a tide-line in a named store.

RENDERING

Source of this paper: *man/man7/pqac.7* in the **freebsd-mac** skill.

```
mandoc -T lint pqac.7
mandoc -T pdf pqac.7 > pqac.pdf
mandoc -T ascii pqac.7 | less
```

`mandoc(1)` emits PDF directly. Install with the rest of the suite manuals under */usr/local/share/man/man7/*.

SEE ALSO

`chmod(1)`, `mandoc(1)`, `mmap(2)`, `mac(4)`, `mac_lomac(4)`, `mac_seeotheruids(4)`, `login.conf(5)`, `mac_grok(7)`, `mac_lomac_grok(7)`, `mac_generic_grok(8)`, `mac_grok(8)`, `mac_lomac_grok(8)`, `sshd(8)`, `mac(9)`

The FreeBSD Handbook ch.19, especially troubleshooting. Fraser, T., "LOMAC: Low Water-Mark Integrity Protection for COTS Environments", USENIX FREENIX 2001 (exceptions for pump, syslogd, sshd are Linux LOMAC, not this module). NIST post-quantum project, for the naming path this paper does not implement.

HISTORY

Spoken first as a refinement of "post-quantum access control" over `mac_lomac(4)` on a classical host: not Kyber, not a quantum OS, a quantum adversary, then the plant, then the same electrons. Written as **pqac** the same day the suite's Known-issues READMEs were finished.

AUTHORS

Grok and the operator of the **freebsd-mac** skills, 2026-08-22. Exoteric practice is the `rc.d` package. Esoteric theses T3 and T4 remain the operator's to prove. Godspeed.